

The log-insertion generating function: $\sum_p \frac{z^p}{p!} (c - \partial_\nu)^p S_\nu = e^{zc} S_{\nu-z}$

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Abstract

Unit 18 of the grammar-paper §4 autoformalisation: the generating function `lem:log_generating`. Summing the log-shift operator against $z^p/p!$ gives $\sum_p \frac{z^p}{p!} \int_0^\infty t^{\nu-1} (c - \log t)^p e^{-\beta t + \beta a \sqrt{t}} dt = e^{zc} S_{\nu-z}(a)$ for $|z| < \nu$ — Taylor’s theorem in the order, $e^{-z\partial_\nu} S_\nu = S_{\nu-z}$. Proved by interchanging the exponential series with the integral. Zero `sorry/axiom`.

1 Setting

This is the last of the log-insertion family (`eq:log_lambda_derivative` $\rightarrow$ `lem:log_shift` $\rightarrow$ this). Where log-shift packaged *one* shifted logarithm as a finite differential operator $(c - \partial_\nu)^p$, the generating function sums them: the whole translation $e^{-z\partial_\nu}$ acts as $\nu \mapsto \nu - z$, and the closed sum is $e^{zc} S_{\nu-z}$.

2 The thing formalised

```
theorem fluctuation_log_generating ( a c z : ℝ ) ( h : 0 < ν ) ( hz : |z| < ν ) :
  HasSum (fun p : ℕ => (z ^ p / p.factorial)
    * t in Set.Ioi 0, (c - Real.log t)^p * t^(-1) * Real.exp (-
  *t + *a*Real.sqrt t))
  (Real.exp (z * c) * fluctuation ( ν - z ) a)
```

3 Proof strategy

The engine is `hasSum_integral_of_dominated_convergence`: from a family F_p with each F_p integrable, a summable pointwise-dominating bound, and a pointwise `HasSum` to f , it concludes `HasSum (∫ F_p) (∫ f)`.

- $F_p(t) = \frac{z^p}{p!} (c - \log t)^p t^{\nu-1} e^{\dots}$; the pointwise limit is $f(t) = e^{z(c-\log t)} t^{\nu-1} e^{\dots}$ (real exponential series `NormedSpace.expSeries_div_hasSum_exp` through `Real.exp_eq_exp_ℝ`, times the constant $t^{\nu-1} e^{\dots}$).
- The bound $\frac{(|z||c-\log t|)^p}{p!} t^{\nu-1} e^{\dots}$ is summable (`Real.summable_pow_div_factorial`) with sum $e^{|z||c-\log t|} t^{\nu-1} e^{\dots}$ (`tsum_mul_right` + the exp series); this dominating function is integrable for $|z| < \nu$, via $|c - \log t| \leq |c| + |\log t|$ and $e^{|z||\log t|} \leq t^{|z|} + t^{-|z|}$, reducing to two fluctuation integrands at orders $\nu \pm |z| > 0$.

- Finally $\int F_p = \frac{z^p}{p!} \int (c - \log t)^p t^{\nu-1} e^{\dots}$ (the log-shift integral) and $\int f = e^{zc} \int t^{\nu-z-1} e^{\dots} = e^{zc} S_{\nu-z}$, using $e^{z(c-\log t)} = e^{zc} t^{-z}$ ($t^{-z} = e^{-z \log t}$, `Real.rpow_def_of_pos`) and $t^{-z} t^{\nu-1} = t^{\nu-z-1}$.

4 Roadblocks and resolutions

Almost every friction was let-binding / beta bookkeeping around the dominated-convergence scaffold:

- The condition $|z| < \nu$ is exactly what keeps $\nu - |z| > 0$, so both dominating fluctuation integrands exist — the same integrability that fails for $\nu \leq |z|$.
- `ring_nf` cannot close an equality of `rpows` with equal-but-unnormalised exponents ($t^{-z} t^{\nu-1} = t^{\nu-z-1}$); `rw` [`← Real.rpow_add`]; `congr 1`; `ring` is the reliable pattern (peel the base, prove the exponent by `ring`).
- `setIntegral_congr_fun` presents its pointwise goal with the target integrand as an *un-beta-reduced* applied lambda, hiding $t^{\nu-z-1}$; writing the `show` with the explicit right-hand side (and `simp only []` to beta-reduce after unfolding the exponential) exposes it.
- Transferring integrability of $\sum'_p$ bound to the closed form uses `Integrable.congr` over the pointwise `htsum`, not a `rw` (the `tsum` pattern does not match syntactically). `abs_add` is now `abs_add_le`.

5 What was Mathlib, what was new

Mathlib supplied the dominated-convergence `HasSum` lemma, the real exponential series and its summability, `tsum_mul_right`, and `Real.rpow_def_of_pos`. New: the $t^{|z|} + t^{-|z|}$ log-exp bound, the integrability of the exponential-of- $|\log|$ dominating function, and the series-integral interchange yielding the closed generating function.

6 Lessons

For a series of integrals converging to an integral, `hasSum_integral_of_dominated_convergence` is far cleaner than `integral_tsum` plus a moment-bound summability argument: the summable bound and its integrable sum are packaged by `tsum_mul_right` + the exp series, and the whole $p!/\nu^p$ moment estimate is avoided — only the single dominating function's integrability (for $|z| < \nu$) is needed.

7 Follow-ups

This completes the entire log-insertion subsection of §4. The sole remaining §4 item is `defn:S_negative/lem:extended_ladder`: extending S_λ to $\lambda \leq 0$ by the recursion $S_\lambda = \frac{\sqrt{\beta}}{2\lambda} b S_{\lambda+1/2}$ and lifting the ladder there — a recursive definition plus analyticity in λ , on which the general- Q number operator rests.